

James Philip Iddon

jamesiddon94@gmail.com • [linkedin](#) • [github.io](#)

Betera, Valencia • +34 640 940 447

Experienced applied physicist / systems engineer with 8+ years in the integration, verification, and operation of large-scale, safety-critical silicon particle detectors and their associated data acquisition systems within large, international collaborations. Proven track record in system-level verification and validation (V&V), requirements-driven development, cross-functional team leadership, and issue resolution under mission-critical conditions.

Skills

Technical: Python • C++ • Hardware-software integration • Verification & Validation (V&V) • CI/CD pipelines • Docker • Bash • Git • Linux • Automated testing and deployment

Systems Engineering: Systems lifecycle development, requirements analysis, integration, validation, deployment, and operational support • Configuration management and quality assurance

Communication & Coordination: Jira • International stakeholder engagement • Cross-functional team leadership • DevOps operations coordination

Code Quality: Code review workflows • Static analysis tools • Testing automation and continuous integration

Professional Experience

Applied Physicist (Severo Ochoa Postdoctoral Fellow)

IFIC / Universitat de Valencia, Valencia, ES

Apr 2026 – Present

Applied Physicist (Staff)

CERN, Geneva, CH

Mar 2025 – Dec 2025

- Accountable for the verification, validation, and integration of the outer barrel of the ATLAS ITK Pixel detector, a large-scale, safety-critical system co-developed by 50 international institutions, illustrating advanced systems engineering and technical coordination capabilities.
- Design of system architecture and V&V protocols to assure the reliability and performance of a 5 billion channel silicon tracker, emphasising expertise in requirements engineering, risk assessment, and quality control.

Applied Physicist (Senior Applied Fellow)

CERN, Geneva, CH

Jul 2022 – Feb 2025

- Managed 24/7 operations and DevOps team for high-reliability sensor system (ATLAS Pixel) supporting particle accelerator data acquisition at LHC.
- Provided on-call operational and DAQ expertise for 24/7 mission-critical operations over 2 years.
- Developed/maintained DAQ software (C++/Python, CI/CD pipelines) for continuous operations in a mission-critical environment.
- Mentored and led 24/7 shift teams drawn from 300-person international pool.
- Chaired international meetings aligning cross-functional teams over 9 time zones.
- Represented ~30 person operations team in 6000+ member international collaboration.
- Served as system operations coordinator for record-setting accelerator year.

Applied Physicist (Doctorant)

University of Liverpool / CERN

Oct 2017 – Jul 2022

- Constructed, characterised, and verified 25% of the Outer Barrel of the ALICE ITS2, the largest monolithic pixel sensor array ever built, with 98% manufacturing yield.
- Developed verification software integrated with production readout and V&V strategies for full detector system (12.5 billion pixels, 10m² silicon).
- Validated performance of complete sensor system against requirements.
- Made first measurement of detection efficiency using 5 million cosmic muon tracks across full detector system.
- Presented technical work in international collaboration meetings (170+ institutions) from small working groups to large plenaries, as well as national and international conferences.

Education

Intensive Spanish Course

Taronja Spanish School, Valencia

Dec 2025 - Apr 2026

PhD in Applied Physics

University of Liverpool / CERN

Oct 2017 – Jul 2022

- Thesis: Construction, Commissioning and Performance Measurements of the Inner Tracking System Upgrade of ALICE at the LHC, focused on advanced systems engineering, integration, and comprehensive verification and validation of complex detector systems.
- Defended June 2022. Shortlisted for the ALICE thesis award.

MPHYS Physics

University of Liverpool

Sep 2013 – Jul 2017

- Integrated masters, specialised in detector physics • Grade: First Class
- Masters project: 'Inner Tracking System Upgrade of the ALICE Experiment at the LHC', characterisation of silicon CMOS MAPS chips.

Publications

All publications as an ALICE and ATLAS author can be found via my Inspire ID: [1618293](#), or via [ORCID](#).

- Total publications and citations are 414 and 7009 respectively with a h-index of 44.

Selected publications:

- ALICE ITS Collaboration [Sensor operating point calibration and monitoring of the ALICE Inner Tracking System during LHC Run 3](#), 2026
- The ATLAS collaboration [Sensor response and radiation damage effects for 3D pixels in the ATLAS IBL Detector](#), Journal of Instrumentation, 2024
- The ALICE collaboration, [ALICE upgrades during the LHC Long Shutdown 2](#), Journal of Instrumentation, 2024
- J. P. Iddon on behalf of the ATLAS collaboration, [Operational experience and performance with the ATLAS Pixel detector at the Large Hadron Collider at CERN](#), Nuclear Instruments and Methods in Physics Research Section A, 2024
- ALICE ITS Collaboration, [First demonstration of in-beam performance of bent Monolithic Active Pixel Sensors](#), Nuclear Instruments and Methods in Physics Research Section A, 2022
- J. P. Iddon on behalf of the ALICE collaboration, [Commissioning of the new ALICE Inner Tracking System](#), Journal of Instrumentation, 2020

Conference Presentations

- Operational Experience and Performance with the ATLAS Pixel detector at the Large Hadron Collider at CERN, HSTD13, Vancouver 2023
- Realising the ALICE ITS Upgrade, IOP Nuclear Colloquium, Remote, Nov 2021
- Performance Measurements from Cosmic Muon Data using the Outer Barrel of the New ALICE Inner Tracking System, PSD12, Birmingham Sept 2021
- Upgrade and Commissioning of the ITS Upgrade of ALICE, INSTR'20, Novosibirsk, Feb 2020
- Entering the precision era of hot QCD measurements, IOP18, Glasgow, Apr 2018

Conference / Workshop Organisation

- Organisation of Operations session, Pixel Week, Milan, 2024
- Organisation of Operations session, Pixel Week, Geneva, 2024
- Organisation of Operations session, Pixel Week, Paris, 2023

Mentorship

- Linus and Marie, Supervisor, High school internship, CERN, 2025
- Zachary Cole Sheridan, Co-supervisor, CERN Summer Student Program, CERN, 2025
- Tim Schloemer, Technical co-supervisor, PhD Qualification Task, CERN, 2024
- Yue Xu, Mentor, Pixel Run Coordination, CERN, 2024
- Kehang Bai, Albert Borbely, Yuan-Tang Chou, Simone Ravera, Sergi Rodriguez Bosca, Khuram Tariq, Elia Schmidt, Mentor, Pixel Run Manager, CERN, 2023 - 2024
- Elia Schmidt, Technical Supervisor, PhD Qualification Task, CERN, 2024
- Lucas Cremer, Technical Supervisor, PhD Qualification Task, CERN (remote), 2023
- Emily Lantham-Taylor, Assisted Supervision, Master's project, Daresbury Laboratory, 2019

Teaching

- ATLAS Run-3 Shift Training, Inner Detector Desk Training, 2024
- PHYS206 Practical Physics II, Teaching Assistant, 2017 - 2019

Leadership

- Pixel DAQ Coordinator 1st Oct 2024 - 31st Jan 2025
- Pixel Run Coordinator 1st Oct 2023 - 30th Sep 2024
- Pixel Deputy Run Coordinator 1st Apr 2023 - 30th Sep 2023
- Leader of the cosmic muon data collection campaign on surface for ALICE ITS2, 2020

Affiliations

- ATLAS, 2022 - present
- ALICE ITS Collaboration 2016 - present
- ALICE, 2016 - 2023
- CERN, 2016 - present
- University of Liverpool 2011 - 2022
- Daresbury Laboratory, 2017 - 2019

One of the 2025 Breakthrough Prize in Fundamental Physics for both ATLAS and ALICE

Full driving licence • Spanish residence permit • English native speaker • Spanish A2+/B1